Abivarsan Ketheeswaran

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Personal profile

Experience in full-stack development, gained through coursework and a significant software project, using agile methodologies. Motivated to learn and contribute, seeking a software Engineer position to enhance technical expertise, collaborate with professionals, and contribute to impactful software solutions.

Education

University of Moratuwa **March 2022 – December 2026**
BSc (Hons) Information Technology and Management *Moratuwa, Sri Lanka*

St. John's College **January 2011 – August 2019**
G.C.E. A/L – Physical Science Stream *Jaffna, Sri Lanka*

Technical Skills

Programming Languages: C#, Java, Python, PHP
Web Technologies: HTML, CSS, JavaScript, TypeScript, Tailwind CSS, Inertia.js, Material UI
Frameworks: ASP.NET Core, Laravel, Spring Boot, FastAPI, React
Databases: MS SQL, MySQL, PostgreSQL, SQLite
Cloud Services: Firebase, Vercel
Version Control: Git, GitHub Actions (CI/CD)
Tools: VS Code, Visual Studio, IntelliJ, GitHub, Postman, Google Gemini API, PHPUnit, Azure DevOps, Figma

Work Experience

Mindox Techno **Mar 2025 – Aug 2025**
Software Engineer Intern *Jaffna, Sri Lanka*

- Gained hands-on experience in semiconductor software systems supporting foundry, memory, and advanced packaging environments.
- Contributed to projects including **CAPCON**, **ASMPTSIM**, and **BSEI** focusing on manual testing, software testing, and host application development.
- Validated system behavior, identified defects, and improved software reliability through structured testing workflows.
- Collaborated with cross-functional R&D teams using task-based development processes.

Projects

Smartwatch Calorie Burn Forecasting & Nutrition Allocation — AI/ML Research| *GitHub* **2026**

- Built a hybrid predictive time-series model for next-day calorie burn forecasting from smartwatch sensor data, achieving $R^2 = 0.9963$ and MAE = 27.08 kcal using clinically validated peer-reviewed TDEE target equations.
- Designed a dynamic inverse-variance ensemble blending strategy with explainability visualization and an automated BMI-driven macronutrient allocation engine, delivering personalized daily dietary targets from wearable data.
- **Technologies:** Python, TensorFlow/Keras, XGBoost, Scikit-learn, SHAP, Pandas, NumPy

Robot Configuration Generator — Industrial Automation Desktop App | *GitHub* **2025**

- Designed and developed a desktop client application utilizing the MVVM architectural pattern for configuring and calibrating industrial robot parameters.
- Implemented software-hardware integration by connecting user interface controls to simulator APIs, enabling dynamic parameter validation prior to hardware deployment.
- **Technologies:** C#, .NET, WPF, MVVM Pattern, Hardware Simulators

Workforce Management System (TaskHive) — Enterprise Web Platform | *GitHub* **2025**

- Architected scalable RESTful Web APIs for an enterprise workforce productivity platform featuring task management, time tracking, and real-time performance analytics.
- Integrated a modern frontend with real-time database updates, implementing repository patterns and dependency injection to establish high code maintainability.
- **Technologies:** C#, ASP.NET Core Web API, MS SQL Server, React (TypeScript), Firebase, Git

Construction Operations & Financial ERP — Enterprise Web Platform | *GitHub* **2026**

- Architected a full-stack operations and financial ERP platform for commercial construction projects, featuring 360° project health tracking, milestone management, and workforce timesheets.
- Engineered real-time financial reconciliation engines calculating Revised Contract Values (RCV), direct incurred costs, gross margins, and client billing ledgers, complemented by 12 exportable operational reports.

- Integrated a high-performance SPA frontend using Inertia.js and React (TypeScript) backed by a serverless deployment on Vercel and Neon PostgreSQL, validated by 85+ automated PHPUnit tests.
- **Technologies:** PHP 8.3, Laravel 11, React 18, TypeScript, Inertia.js, PostgreSQL (Neon), Tailwind CSS, Vite, Vercel, PHPUnit, Git

Autonomous AI Job Search & Evaluation Agent — Full-Stack & Cloud Automation | *GitHub* **2026**

- Engineered an automated ETL pipeline that scrapes, normalizes, and filters software engineering listings across 5 job portals (LinkedIn, TopJobs, ITPro, Rooster, Xpress) with strict workplace and location heuristics.
- Integrated Google Gemini LLM with structured prompt engineering to evaluate candidate compatibility (matching/missing skills analysis) and automated 3-hour recurring sweeps via GitHub Actions with instant Telegram alerts.
- Built a mobile-responsive web dashboard and offline PWA backed by a local FastAPI/SQLite repository pattern and Vercel static hosting, featuring 1-click application tracking and dynamic filtering.
- **Technologies:** Python, FastAPI, SQLite, Google Gemini API, GitHub Actions (CI/CD), Telegram Bot API, JavaScript, Tailwind CSS, Service Workers (PWA), Vercel, Git

Uni Tutor Platform — Multi-Tier Web Platform | *Back-end* | *Front-end* **2024**

- Engineered backend APIs to power a university tutoring marketplace connecting students and lecturers, featuring chat, search filtering, review systems, and virtual coin transactions.
- Designed and implemented an Admin Verification Dashboard featuring tutor identity verification via university ID checks, user suspension/restoration logic, report generation, and data analytics dashboards.
- **Technologies:** C#, ASP.NET Core, MS SQL Server, React (TypeScript), Firebase

TechHub E-Commerce Platform — Microservices Architecture | *Back-end* | *Front-end* **2024**

- Co-developed a distributed microservices e-commerce system in a team of six, designing and building the core Order Management Service covering order placement, status tracking, and payment workflows.
- Configured event-driven notification services, integrated Stripe online payment APIs, configured API Gateway routing, and containerized microservices for decoupled deployment.
- **Technologies:** Java, Spring Boot, React.js, MySQL, Apache Kafka, Docker, Stripe API, Postman

Employee Management Application — Enterprise HR & IT System | *GitHub* **2024**

- Developed an enterprise workforce administration platform featuring role-based access control with distinct System Administrator and Employee dashboards.
- Built an Executive Approval Center for reviewing leave and overtime requests, an automated working-days calculation engine excluding holidays, and 6-digit SMTP OTP email verification for registration and password recovery.
- **Technologies:** C#, .NET 8, ASP.NET Core MVC, Entity Framework Core 9, MS SQL Server, System.Net.Mail (SMTP), Bootstrap 5

Top Speed Automobile Showcase & Fleet Platform — Automotive Systems | *GitHub* **2024**

- Engineered an automotive fleet management platform supporting full vehicle technical dossiers, dynamic category filtering, and side-by-side performance spec comparisons with automatic winner badge allocation.
- Built an interactive analytics dashboard for fleet metrics, a test-drive reservation concierge with automated email dispatch, role-based access control, and single-click CSV data export endpoints.
- **Technologies:** C#, .NET 8, ASP.NET Core MVC, Entity Framework Core 8, MS SQL Server, ASP.NET Identity, Chart.js, MailKit/MimeKit (SMTP), DataTables

Task Manager Application — Interactive IT Workflow App | *GitHub* **2024**

- Built a productivity platform featuring an interactive drag-and-drop HTML5 Kanban board with real-time status synchronization, step-by-step subtask checklist builders, and glowing priority badges.
- Integrated automated email digest notifications summarizing pending tasks, header overdue alerts, multi-parameter search filtering, single-click CSV metadata export, and print-ready PDF reporting.
- **Technologies:** C#, .NET 8, ASP.NET Core MVC, Entity Framework Core 8, MS SQL Server, System.Net.Mail (SMTP), AJAX, Vanilla JS

QA Automation Test Suite for Web Applications — BDD Testing | *GitHub* **2025**

- Built automated Behavior-Driven Development (BDD) test suites covering end-to-end user workflows, API integration endpoints, UI element interaction validation, and regression test scenarios.
- **Technologies:** Playwright, Cucumber, JavaScript, TypeScript, UI & API Automation Testing

Soft Skills

- Verbal Communication
- Problem Solving
- Attention to Detail
- Responsiveness
- Team Work

Languages

- English(Professional Working Proficiency)
- Tamil(Native)

Interests

- Playing Football
- Browsing Internet

Extracurricular Activities

Secretary, Science Union 2018, St. John's College, Jaffna
Secretary, St. John's Ambulance 2018, St. John's College, Jaffna
Industry Visit and Organizing Team Building Event, Organizing Committee 2023, Creative Software
Participation, HackMoral 6.0 2024, INTECS, Batch21, FIT, UOM
Participation, Monthly Code Surge 2024, Pagero
Participation, AlgoXplore 1.0 2024, NSBM Green University
Representative, Final Year, ITM, University of Moratuwa

References

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